

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000488 - Software Metrics

DEGREE PROGRAMME

10AM - Master Universitario En Ingenieria Del Software

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	103000488 - Software Metrics
No of credits	4 ECTS
Type	Compulsory
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	10AM - Master Universitario en Ingeniería del Software
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Tomas San Feliu Gilabert (Subject coordinator)	D5106	tomas.sanfeliu@upm.es	M - 10:00 - 13:00 W - 10:00 - 13:00
Jose Antonio Calvo- Manzano Villalon	D5107	joseantonio.calvomanzano@upm.es	M - 10:30 - 13:30 W - 10:30 - 13:30

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CE10 - Evaluar de forma objetiva los procesos y productos frente a los estándares y normas aplicables.

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio (RD)

CG12 - Comprensión amplia de las técnicas y métodos aplicables en una especialización concreta, así como de sus límites

CG18 - Capacidad de trabajar y comunicarse también en contextos internacionales

CG2 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios (RD)

3.2. Learning outcomes

RA79 - The student will be able to analyze data for estimation, planning and quality control in software projects

RA78 - The student will be able to lead and implement measurement plans for the evaluation of processes and products

RA45 - Es capaz de analizar datos para la estimación, planificación y control de calidad en proyectos software

RA44 - Es capaz de liderar e implantar planes de medida para la evaluación de procesos y productos

RA11 - Understands the interrelation between product quality and process quality

RA16 - The student will be able to evaluate any software system design.

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

The students will:

- Understand the theoretical aspects of software measurements.
- Demonstrate the knowledge of software metrics.
- Demonstrate the knowledge of using software metrics in software development, software maintenance, and software project management.
- Demonstrate the knowledge of developing and maintaining a measurement program.

4.2. Syllabus

1. Introduction to measurement theory
 - 1.1. Why measure?
 - 1.2. Measure elements
2. Measuring Product Attributes
 - 2.1. Internal Product Attributes
 - 2.2. External Product Attributes
3. Measuring Process
4. Designing Metrics Program
 - 4.1. Goal Question Metrics
5. Frameworks for Software Measurements
 - 5.1. ISO 15939
 - 5.2. Practical Software and System Measurement
 - 5.3. Continuous Iterative Development
6. Agile Metrics
7. Visualization and Decision Making with Software Measurements

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Chapter 1.1 Why Measure Duration: 02:00 Lecture			
2	Chapter 1.2: Measuring Elements Duration: 02:00 Lecture			
3	Chapter 2.1 Measuring Internal Product Attributes Duration: 02:00 Lecture			Quiz 1 Online test Progressive assessment Not Presential Duration: 00:30
4	Chapter 2.2 Measuring External Product Attributes Duration: 02:00 Lecture			
5	Chapter 2.2 Measuring External Product Attributes Duration: 02:00 Lecture			
6	Chapter 2.2 Measuring External Product Attributes Duration: 02:00 Lecture			Quiz 2 Online test Progressive assessment Not Presential Duration: 00:30
7	Chapter 4. Designing Metrics Program Duration: 02:00 Lecture			
8	Chapter 4. Designing Metrics Program Duration: 02:00 Lecture			
9	Chapter 4. Designing Metrics Program Duration: 02:00 Lecture			Quiz 3 Online test Progressive assessment Not Presential Duration: 00:30
10	Chapter 4. Designing Metrics Program Duration: 02:00 Lecture			Measurement Goal Definition Group work Progressive assessment Not Presential Duration: 02:00
11	Chapter 4. Designing Metrics Program Duration: 02:00 Lecture			Quiz 4 Online test Progressive assessment Not Presential Duration: 00:30

12	Chapter 5. Frameworks for Software Measurements Duration: 02:00 Lecture			
13	Chapter 6. Agile Metrics Duration: 02:00 Lecture			Quiz5 Online test Progressive assessment Not Presential Duration: 00:30
14	Chapter 7 Visualization and Decision Making with Software Measurements Duration: 02:00 Lecture			
15	Evaluation Recovery Test Duration: 02:00 Additional activities			Final Report Group work Progressive assessment Not Presential Duration: 02:00 Evaluation Recovery Test Online test Progressive assessment Presential Duration: 02:00
16				Progressive Evaluation Online test Progressive assessment Not Presential Duration: 02:00 Global Evaluation Online test Global examination Not Presential Duration: 02:00
17				

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Quiz 1	Online test	No Presential	00:30	5%	0 / 10	CE10 CG2 CG18
6	Quiz 2	Online test	No Presential	00:30	5%	4 / 10	CE10
9	Quiz 3	Online test	No Presential	00:30	5%	4 / 10	CE10 CG2 CG18
10	Measurement Goal Definition	Group work	No Presential	02:00	20%	4 / 10	CE10 CG12
11	Quiz 4	Online test	No Presential	00:30	5%	4 / 10	CE10 CG2 CG18
13	Quiz5	Online test	No Presential	00:30	5%	0 / 10	CE10 CG2 CG18
15	Final Report	Group work	No Presential	02:00	25%	4 / 10	CG2 CG12 CG18 CG1
15	Evaluation Recovery Test	Online test	Face-to-face	02:00	%	5 / 10	CG12 CG18 CG1 CE10 CG2
16	Progressive Evaluation	Online test	No Presential	02:00	30%	5 / 10	CG2 CG12 CG18 CG1 CE10

6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
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16	Global Evaluation	Online test	No Presential	02:00	100%	5 / 10	CE10 CG2 CG12 CG18 CG1
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6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Extraordinary Evaluation	Written test	Face-to-face	00:30	100%	5 / 10	CE10 CG2 CG18

6.2. Assessment criteria

The final grade for the course is determined based on the following components:

Five questionnaires: 25%

Measurement Goal Definition: 20%

Final report: 25%

Progressive Evaluation questionnaire: 30%

To pass each component, students must achieve a minimum score of 5.0 out of 10. If a student does not meet this threshold for any component (but scores above zero), they may retake the respective component during the Evaluation Recovery Test in Week 15. The new scores from the Evaluation Recovery Test will replace the original scores and will be combined with the marks from other components (including group exercises) to calculate the grade for the ordinary evaluation period.

In Week 16, two exams will take place:

- **Progressive Evaluation Exam (30% of the final grade):** This exam is for students who have successfully completed all the aforementioned components. The score from this exam will complete the remaining 30% of the final grade, bringing the total to 100% of the course grade.
- **Global Evaluation Exam:** This exam is for students who have not passed one or more of the progressive evaluation components. The score from this exam will replace the failed components and will be combined with the marks from other components (including group exercises and class participation) to calculate the final grade for the ordinary evaluation period.

If a student does not pass the course during the ordinary evaluation period, they may take a final comprehensive test during the extraordinary evaluation period in July. The grade from this test will be used to calculate the final course grade.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Metrics and Models in Software Quality Engineering	Bibliography	Stephen Kan, Metrics and Models in Software Quality Engineering, Addison Wesley 2003
Measuring the Software Process	Bibliography	Anita Carleton, Measuring the Software Process, Addison Wesley
The big book of Six Sigma training games	Bibliography	Chris Chen and Hadley Roth, The big book of Six Sigma training games, McGraw-Hill, 2005

8. Other information

8.1. Other information about the subject

All student work is required to be their own. While we encourage research and reuse, all submissions must be your own, original creation.

Copying/ Plagiarism is NOT tolerated. This is standard policy.

This applies to ALL work, be it code related or written/ essay style submission.

Use of a GPT/ AI, you may use GPT or other AI tools for searching and/ or reference, with the following conditions:

1.- If you use a GPT (ChatGPT or other, similar tools) you must also do the following:

- Submit your prompts for GPT (as a separate file) as part of your work
- Provide a confirming statement that GPT was used as reference only, and your work is original.

2.-If ANY of your work is found to be copied (from online sources OR ChatGPT/ AI), you would be subject to serious penalties which could be one or more of the following:

- Zero on the assignment
- Zero on the final grade

Regarding innovative teaching methodologies:

In the Software Metrics course, students are required to create a project that addresses specific requirements by forming teams of at least three to eight individuals (practical approach). This is in reference to innovative teaching methodologies.

The project is selected by the groups from a collection of project statements that represent various companies with distinct challenges that they must address. To finalize this project, it is necessary to implement the procedures for generating the measurement program. The instructor initially provides a description of the project and its scope, as well as the objectives that must be met in each development step. The instructor provides a detailed explanation of

the various activities that are to be completed, ensuring that the teams are aware of the anticipated deliverables.

The endeavor must be planned by the teams, necessitating a strong commitment from all members. The teams are required to implement the theory in the project as it is explicated. The instructor is available to address any inquiries or concerns at any time.

This course employs a variety of innovative teaching methodologies to reinforce and motivate student learning.

1) The Virtual Collaboration methodology is implemented throughout the undertaking. Students can either complete the various activities in person or collaborate using applications such as Teams or Zoom to communicate and OneDrive to share files. They are divided into teams. This enables them to collaborate remotely. Additionally, the instructor has the option to attend any of these meetings in order to address any queries or concerns.

2) The Learning-by-Doing methodology is applied in the course through a project developed in ten consecutive steps. The project, developed in ten consecutive stages, serves as the vehicle through which the Learning-by-Doing methodology is implemented in the course. To accomplish this, the instructor initially provides practical examples and theoretical explanations of the various activities, which are then executed by the teams.